Schur Rings, Dihedral Groups, and Relative Difference Sets

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Schur Rings

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Definition - Schur Rings (Wielandt 1949)

Let $\mathcal{D} = \{C_1, C_2, \ldots, C_r\}$ be a partition of G and let $\mathfrak{S}$ be the subspace of $\mathbb{Q}[G]$ spanned by $C_1, C_2, \ldots, C_r$. We say $\mathfrak{S}$ is a **Schur Ring** (or S-ring) if

- $\{1\} \in \mathcal{D}$,
- $C^{(-1)} \in \mathcal{D}$, for all $C \in \mathcal{D}$,
- $CD = \sum_{E \in \mathcal{D}} \lambda_E E$, for all $C, D \in \mathcal{D}$.

Examples of Schur Rings

Examples of Schur Rings:

- Every group algebra $\mathbb{Q}[G]$ is itself a Schur ring with partition $G = \displaystyle\bigcup_{g\in G} \{g\}$, called the **discrete Schur ring**.

  e.g. $G = \mathcal{Z}_7 = \langle z \rangle$:

$$\mathfrak{S} = \text{Span}\{\ 1,\ z,\ z^2,\ z^3,\ z^4,\ z^5,\ z^6\}$$

Examples of Schur Rings

Examples of Schur Rings:

- Let $\mathcal{D} = \{\{1\}, G \setminus \{1\}\}$. The associated Schur ring is known as the **Trivial Schur ring**.

  e.g. $G = \mathcal{Z}_7 = \langle z \rangle$:

$$\mathfrak{S} = \text{Span}\{\ 1,\ z + z^2 + z^3 + z^4 + z^5 + z^6\ \}$$

Examples of Schur Rings

Examples of Schur Rings:

- Let $\mathcal{H} \leq \text{Aut}(G)$. Let $G^{\mathcal{H}}$ denote the Schur ring associated to the partition of G afforded by the action of $\mathcal{H}$ on G via automorphisms, called an **automorphic Schur ring**.

  e.g. $G = \mathcal{Z}_7 = \langle z \rangle$, $\mathcal{H} = \langle 2 \rangle$:

$$\mathfrak{S} = \text{Span}\{\ 1,\ z + z^2 + z^4,\ z^3 + z^5 + z^6\ \}$$

Examples of Schur Rings

Examples of Schur Rings:

- Let $H \times K = G$. Let $\mathfrak{S}$ and $\mathfrak{T}$ be Schur rings over H and K, respectively. Then $\mathfrak{S} \times \mathfrak{T}$ is a Schur ring over G, called the **Direct Product** of $\mathfrak{S}$ and $\mathfrak{T}$. Its partition is constructed by all the possible products of sets from the partitions of $\mathfrak{S}$ and $\mathfrak{T}$.

  e.g. $G = \mathcal{Z}_6 = \langle z \rangle$, $H = \mathcal{Z}_2$, $K = \mathcal{Z}_3$:

  $$\mathfrak{S} = \mathrm{Span}\{\ 1,\ z^3,\ z^2 + z^4,\ z + z^5\ \}$$

Examples of Schur Rings

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- Let $H \trianglelefteq G$. Let $\mathfrak{S}$ and $\mathfrak{T}$ be Schur rings over H and G/H, respectively. The partition which affords $\mathfrak{T}$ over G/H can be inflated into a partition of G. The amalgamation of the partitions of $\mathfrak{S}$ and $\mathfrak{T}$ affords a Schur ring, $\mathfrak{S} \wedge \mathfrak{T}$, over G, called the **Wedge Product** of $\mathfrak{S}$ and $\mathfrak{T}$.

  e.g. $G = \mathcal{Z}_6 = \langle z \rangle$, $H = \mathcal{Z}_2$, $K = \mathcal{Z}_3$:

$$\mathfrak{S} = \mathrm{Span}\{\ 1,\ z^2,\ z^4,\ z + z^3 + z^5\ \}$$

Traditional Schur Rings

The four types of Schur rings discussed so far:

- trivial
- automorphic
- direct products
- wedge products

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Traditional Schur Rings

Theorem - Bastian, Brewer, Humphries, M, Thompson (2020)

All Schur rings over the following groups are traditional:

- The Infinite Cyclic Group $\mathcal{Z}$
- The Virtually Cyclic Group $\mathcal{Z} \times \mathcal{Z}_2$
- Any Torsion-Free Locally Cyclic Group, such as $\mathbb{Q}$
- The Virtually Locally Cyclic Group $G \times \mathcal{Z}_2$, where G is a torsion-free locally cyclic group

Traditional Schur Rings

Theorem - Chen, He (2021)

All Schur rings over the virtually cyclic group $\mathcal{Z} \times \mathcal{Z}_3$ are traditional.

A prime p is **Fermat** if $p = 2^k + 1$. There are probably only five such primes.

A prime p is **safe** if $p = 2q + 1$ and q is itself prime (a Sophie Germain prime). There are probably infinitely many safe primes.

Theorem - Bastian, M (To Appear)

All Schur rings over the virtually cyclic group $\mathcal{Z} \times \mathcal{Z}_p$ are traditional, where p is a Fermat or safe prime.

Non-Traditional Schur Rings

Traditional Schur rings are the best known Schur rings and the easiest to construct, but many finite groups are known to have non-traditional Schur rings.

Theorem - Humphries (2020)

There exists non-traditional Schur rings over surface groups, some free products with amalgamation, $\mathrm{PSL}(2,\mathbb{Z})$, and some groups related to crystallographic groups.

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There exists non-traditional Schur rings over dihedral groups.

This Schur ring is constructed using a difference set.

Difference Sets

Definition - Difference Sets (Bruck 1955)

Let H be a finite group, with $|H| = v$. Let $D \subseteq H$, with $|D| = k$. We say that D is a $(v, k, \lambda)-$**difference set** of H if each non-identity element $h \in H$ can be expressed as $h = gk^{-1}$, for $g, k \in D$, in exactly λ distinct way.

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D is a difference set if and only if $DD^{(-1)} = k + \lambda(H - 1)$

Difference Sets

Examples of Difference Sets:

- $\emptyset$, H
- $\{h\}$, $H - h$

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- $\emptyset$, H
- $\{h\}$, $H - h$
- $H = \mathcal{Z}_7 = \langle z \rangle$, $D = \{z, z^2, z^4\}$:

	z^{-1}	z^{-2}	z^{-4}
z	1	z^6	z^4
z^2	z	1	z^5
z^4	z^3	z^2	1

The Difference-Set Schur Ring

Let H be a finite abelian group. Let D be a difference set over H. Let

$$G = \mathrm{Dih}(H) = H \rtimes \mathcal{Z}_2 = \langle H, s\rangle$$

be the dihedral group over H. Then

$$\mathfrak{S} = \mathrm{Span}\{1, H-1, Ds, Hs - Ds\} = \mathrm{Span}\{1, H, Ds, Hs\}$$

is a rank-4 commutative Schur Ring, the so-called **Difference-Set Schur Ring**.

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	1	H	Ds	Hs
1	1	H	Ds	Hs
H	H	vH	kH	vHs
Ds	Ds	kH	$k+\lambda(H-1)$	kH
Hs	Hs	kHs	kH	vH

Schur Rings over D_n

Let $D_n = \mathrm{Dih}(\mathcal{Z}_n) = \langle r, s \mid r^n = 1, s^2 = 1, sr = r^{-1}s \rangle$.

Theorem - Shiu (1989)

All Schur rings over D_7 are traditional or are DS Schur rings.

For $G = D_7$,

$$\mathfrak{S} = \mathrm{Span}\{1, r + r^2 + r^3 + r^4 + r^5 + r^6, rs + r^2s + r^4s, s + r^3s + r^5s + r^6s\}.$$

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Theorem - M (2023)

Let p be a Fermat or safe prime. Then all Schur rings over D_p are traditional or are DS Schur rings.

DS Schur Rings over Semidirect Products

We generalize the previous construction.

Theorem - M (2023)

Let H be an abelian group, and let K be a group. Let $\varphi : K \to \mathcal{Z}_2$ be surjective. Let $G = H \rtimes_\varphi K$. Let D be a difference set of H. Then

$$\mathfrak{S} = \text{Span}(\{x, (H-1)x \mid x \in K, \varphi(x) = 1\} \cup \{Dx, (H-D)x \mid x \in K, \varphi(x) = -1\})$$

is a Schur ring over G.

DS Schur Ring over Dicyclic Group

Let $H = \mathcal{Z}_7 = \langle r \rangle$, $K = \mathcal{Z}_4 = \langle s \rangle$, and let

$$G = H \rtimes K = \langle r, s \mid r^7 = 1, s^4 = 1, sr = r^{-1}s \rangle$$

be the dicyclic group of order 28.

Then

$$\begin{aligned}\mathfrak{S} = \mathrm{Span}(&1, r + r^2 + r^3 + r^4 + r^5 + r^6, \\ &rs + r^2s + r^4s, s + r^3s + r^5s + r^6s, \\ &s^2, rs^2 + r^2s^2 + r^3s^2 + r^4s^2 + r^5s^2 + r^6s^2, \\ &rs^3 + r^2s^3 + r^4s^3, s^3 + r^3s^3 + r^5s^3 + r^6s^3)\end{aligned}$$

is a DS Schur ring over G.

Relative Difference Sets

Definition - Relative Difference Sets (Butson 1963)

Let H be a finite group, with $|H| = mn$. Let $N \trianglelefteq H$, with $|N| = n$. Let $D \subseteq H$, with $|D| = k$. We say that D is a $(m, n, k, \lambda)-$**relative difference set** of H relative to n if for each element $h \in H \setminus N$ can be expressed as $h = gk^{-1}$, for $g, k \in D$, in exactly λ distinct way.

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D is a relative difference set if and only if $DD^{(-1)} = k + \lambda(H - N)$

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A relative difference set is **semiregular** if $k = \lambda n$. This implies that $DN = H$.

Relative Difference Sets

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- no semiregular relative difference set over a cyclic group

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Examples of Semiregular Relative Difference Sets:

- no semiregular relative difference set over a cyclic group
- $H = \mathcal{Z}_3^2 = \langle a, b\rangle$, $N = \langle a\rangle$, $D = \{1, ab, ab^2\}$:

	1	a^2b^2	a^2b
1	1	ab	ab^2
ab	ab	1	b^2
ab^2	ab^2	b	1

The Relative Difference-Set Schur Ring

Theorem - M (2023)

Let H be an abelian group with normal subgroup N. Let $G = \mathrm{Dih}(H)$. Let D be a semiregular relative difference set of H relative to N. Then

$$\mathfrak{S} = \mathrm{Span}\{1, N-1, H-N, Ds, Hs-Ds\} = \mathrm{Span}\{1, N, H, Ds, Hs\}$$

is a rank-5 commutative Schur Ring, the so-called **Relative-Difference-Set Schur Ring**.

$$(Ds)(Ds) = DD^{(-1)}(ss) = DD^{(-1)} = k + \lambda(H-N)$$

$$N(Ds) = (ND)s = Hs$$

The Relative-Difference-Set Schur Ring

	1	N	H	Ds	Hs
1	1	N	H	Ds	Hs
N	N	nN	nH	Hs	nHs
H	H	nHs	mnH	kHs	$mnHs$
Ds	Ds	H	kH	$k+\lambda(H-N)$	kH
Hs	Hs	nHs	kHs	kH	mnH

RDS Schur Ring over $\mathrm{Dih}(\mathcal{Z}_3^2)$

For

$$\begin{aligned} G &= \mathrm{Dih}(\mathcal{Z}_3^2) \\ &= \langle a, b, s \mid a^3 = b^3 = s^2 = 1, ab = ba, sa = a^2s, sb = b^2s \rangle, \end{aligned}$$

we have

$$\begin{aligned} \mathfrak{S} = \mathrm{Span}\{&1, a + a^2, b + b^2 + ab + a^2b^2 + ab^2 + a^2b, \\ &s + abs + ab^2s, as + a^2s + bs + b^2s + a^2b^2s + a^2bs\}. \end{aligned}$$

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Let H be an abelian group with normal subgroup N, and let K be a group. Let $\varphi : K \to \mathcal{Z}_2$ be surjective. Let $G = H \rtimes_\varphi K$. Let D be a semiregular relative difference set of H relative to N. Then

$$\begin{aligned}\mathfrak{S} = \operatorname{Span}(x, (N-1)x, (H-N)x \mid x \in K, \varphi(x) = 1\} \\ \cup \{Dx, (H-D)x \mid x \in K, \varphi(x) = -1\})\end{aligned}$$

is a Schur ring over G.

Acknowledgments

Thank you for listening!